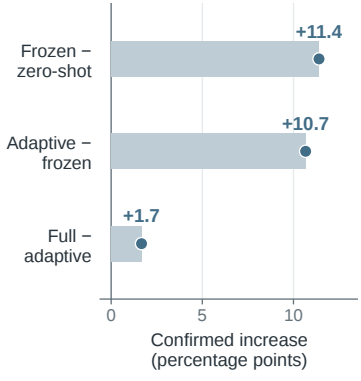
a **Evidence-acquisition stages and repeat stability**

Topic	Zero-shot	Frozen retrieval	Adaptive search	Full	Full range (%)	Same label (%)
DBPs	26.3 ± 0.6	15.7 ± 1.2	43.7 ± 3.2	**46.7 ± 3.2**	43–49	76.0
Blood exposome	3.0 ± 0.0	14.3 ± 2.1	18.7 ± 1.2	**18.7 ± 3.1**	16–22	90.0
Drugs	42.7 ± 0.6	57.7 ± 1.5	58.3 ± 2.5	**58.7 ± 2.1**	57–61	77.0
Endogenous	23.7 ± 0.6	20.0 ± 2.6	26.7 ± 4.2	**30.0 ± 1.7**	28–31	83.0
Microbial NP	10.0 ± 0.0	55.0 ± 2.0	68.7 ± 5.1	**70.3 ± 2.1**	68–72	84.0
All topics	**21.1**	**32.5**	**43.2**	**44.9**		

b **Gains between stages**



c **Critique-model comparison**

Topic	Critique	Adaptive	Full	Change (pp)
DBPs	GPT	43.7 ± 3.2	46.7 ± 3.2	+3.0
	Gemini	39.7 ± 7.6	42.3 ± 6.5	+2.7
Blood exposome	GPT	18.7 ± 1.2	18.7 ± 3.1	+0.0
	Gemini	16.0 ± 4.4	17.7 ± 2.1	+1.7
Drugs	GPT	58.3 ± 2.5	58.7 ± 2.1	+0.3
	Gemini	58.7 ± 4.9	59.7 ± 3.2	+1.0
Endogenous	GPT	26.7 ± 4.2	30.0 ± 1.7	+3.3
	Gemini	25.0 ± 4.0	27.0 ± 3.6	+2.0
Microbial NP	GPT	68.7 ± 5.1	70.3 ± 2.1	+1.7
	Gemini	67.0 ± 2.6	69.0 ± 2.0	+2.0

d **Feedback mechanisms and research activity**

Critique feedback	Confirmed (%)	Rounds	Tokens	Cost (US$)	New files
A Criterion-directed	**58.0**	1.62	209,138	0.0880	148.9
B General review	**48.0**	1.08	172,190	0.0696	67.9
C No explicit rubric	**51.0**	1.07	148,995	0.0613	74.7